Capacitive touch

DETECT | RASPBERRY PI | INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses



Touch lamp

DETECT: Finger touches the pad → DECIDE: Is the lamp already on? → DO: Toggle the light



Banana piano

DETECT: A banana is touched → DECIDE: Which note should play? → DO: Play the note



Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import DigitalInputDevice

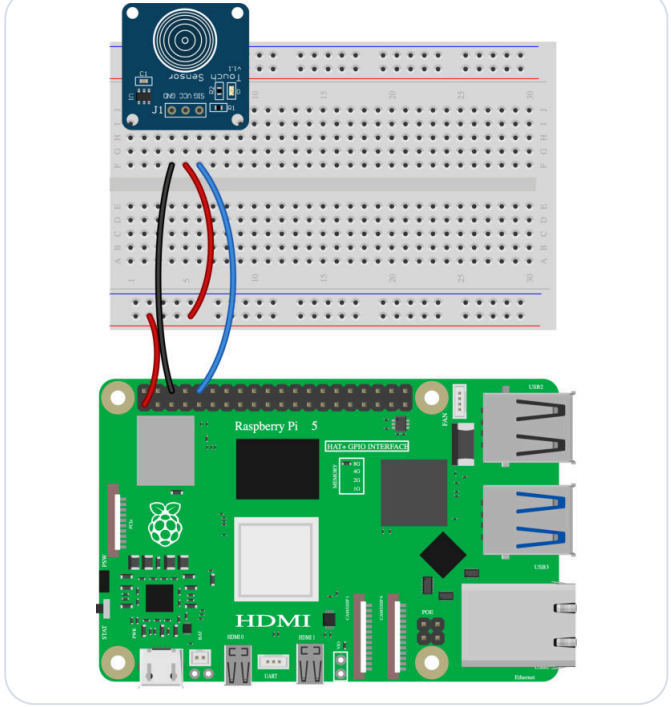
touch = DigitalInputDevice(15, pull_up=False)

touched = touch.is_active
```

The TTP223 output goes high while touched. `touched` is True or False.

Wiring

RASPBERRY PI 5



GND (pin 6) → GND, **3V3** (pin 1) → VCC, **GPIO 15** (pin 10) → SIG